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Modular pipe loader assembly

Abstract

A horizontal directional drilling machine having a modular pipe loader system. The system comprises a first and second pipe loader assembly supported on a drill frame. Each assembly supports a shuttle arm. The shuttle arms are configured to move independently of one another along a shuttle path that is traverse to a longitudinal axis of the drill frame. Movement of each shuttle arm is powered by an actuator supported on each pipe loader assembly. Each pipe loader assembly includes a sensor used to measure parameters related to the position of each shuttle arm relative to the drill frame. A controller analyzes the measured parameters and directs operation of each actuator in order to keep the shuttle arms moving in unison during operation.

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Background/Summary

(1) The present disclosure is directed to an apparatus comprising an elongate frame having a longitudinal axis. The apparatus also comprises a first shuttle arm supported by the frame and movable along a first shuttle path transverse to the longitudinal axis of the frame, and a second shuttle arm supported by the frame and movable along a second shuttle path spaced from, but parallel to, the first shuttle path. The apparatus also comprises a first actuator configured to power movement of the first shuttle arm along the first shuttle path, and a second actuator configured to power movement of the second shuttle arm along the second shuttle path, independent of the first actuator.

(2) The apparatus further comprises a first sensor that periodically measures a first parameter that is either the position of the first shuttle arm or a parameter from which such position may be calculated, and a second sensor that periodically measures a second parameter that is either the position of the second shuttle arm or a parameter from which such position may be calculated. The apparatus even further comprises a controller in communication with the first and second sensors and with the first and second actuators. The controller is configured to evaluate the first and second parameters, and to issue commands to one or both of the first and second actuators in response to that evaluation.

(3) The present disclosure is also directed to a method of using an apparatus. The apparatus comprises an elongate frame having a longitudinal frame axis, a first shuttle arm supported by the frame and movable along a first shuttle path traverse to the frame axis, and a second shuttle arm supported by the frame and movable along a second shuttle path spaced from, but parallel to, the first shuttle path. The method comprises the step of moving each of the first and second shuttle arms relative to the frame, and determining the velocity of each of the first and second shuttle arms at successive positions along their respective shuttle paths. The method further comprises the step of modifying the velocity of one or more shuttle arms in response to the determinations of velocity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a horizontal directional drilling system.

(2) FIG. 2 is a right side elevational view of a drilling machine having a modular pipe loading system.

- (3) FIG. 3 is a left side perspective view of a portion of the drilling machine shown in FIG. 2. Various components of the drilling machine shown in FIG. 2 have been removed to better view the displayed portion of the drilling machine.
- (4) FIG. 4 is a top plan view of the modular pipe loading system shown in FIG. 2. The system is shown supported on a drill frame.
- (5) FIG. 5 is right side elevational view of the portion of the drilling machine shown in FIG. 3.
- (6) FIG. 6 is a right side elevational view of a second pipe loader assembly used with the modular pipe loading system shown in FIG. 2.
- (7) FIG. 7 is a bottom perspective view of the second pipe loader assembly shown in FIG. 6.
- (8) FIG. 8 is a left side elevational view of a first shuttle arm supported on a first pipe loader assembly used with the modular pipe loading system shown in FIG. 2. A portion of the first pipe loader assembly has been removed to expose a first sensor.
- (9) FIG. 9 is a bottom plan view of the first pipe loader assembly used with the modular pipe loading system shown in FIG. 2.
- (10) FIG. 10 is a bottom perspective view of a rearward end of the pipe loader assembly shown in FIG. 9.
- (11) FIG. 11 is a front perspective view of the second pipe loader assembly shown in FIG. 6.
- (12) FIG. 12 is a left side perspective view of the first pipe loader assembly shown in FIG. 9. The first lift assembly has been removed to expose the first sensor.
- (13) FIG. 13 is a bottom plan view of the second pipe loader assembly shown in FIG. 6, using an alternative embodiment of a sensor.
- (14) FIG. 14 is a right side elevational view of the second pipe loader assembly shown in FIG. 13. Portions of the assembly and sensor have been removed to expose the sensor.
- (15) FIG. 15 is a flow chart depicting a method for re-aligning misaligned shuttle arms.
- (16) FIG. 16 is a flow chart depicting a method for preventing the shuttle arms from becoming misaligned.
- (17) FIG. 17 is a flow chart depicting a method of using the shuttle arms independently while making up a drill string.
- (18) FIG. 18 is a flow chart depicting a method of using the shuttle arms independently while removing pipe sections from the drill string.
- (19) FIG. 19 is a flow chart depicting a method of using the shuttle arms independently while preparing the drilling machine for transport.

DESCRIPTION

- (20) Turning now to the figures, FIG. 1 shows a drilling machine 10 sitting on a ground surface 12. The drilling machine 10 is configured for use in a “horizontal boring” or “horizontal directional drilling” operation. The drilling machine 10 is used to create a horizontal borehole 14 below the ground surface 12. The borehole 14 provides space underground for installation of a utility pipeline.
- (21) Extending from the drilling machine 10 is a drill string 16. The drill string 16 is made up of a plurality of pipe sections 18 attached end-to-end. The drill string 16 is connected to a downhole tool 20 at its first end 22 and the drilling machine 10 at its second end 24.
- (22) The downhole tool 20 comprises a drill bit 26 and a beacon contained within a beacon housing 28. In operation, the drill bit 26 bores underground and advances the downhole tool 20 and the drill string 16 forward, thereby creating the borehole 14. The drilling machine 10 adds the plurality of pipe sections 18 to the drill string 16 as the downhole tool 20 advances underground. An above-ground tracker 30 tracks a signal emitted from the beacon during operation.
- (23) Turning to FIGS. 2 and 3, the drilling machine 10 comprises an operator station 32, engine compartment 34, and an elongate drill frame 36 supported on a pair of endless tracks 38. The drill frame 36 has a longitudinal axis 40, as shown in FIG. 3. The drill frame 36 supports a carriage 42 at its first end 44 and a pair of wrenches 46 at its second end 48.

(24) The drill frame **36** further supports a modular pipe loader assembly **51**. The modular pipe loader assembly **51** comprises a first and second pipe loader assembly **50** and **52**. As will be described later herein, the first and second pipe loader assemblies **50** and **52** are configured to operate independently of one another.

(25) Continuing with FIGS. **2** and **3**, the pipe loader assemblies **50** and **52** support a pipe box **54** housing pipe sections **18**. The pipe loader assemblies **50** and **52** and the pipe box **54** are supported adjacent to the drill frame **36** and between the carriage **42** and wrenches **46**. The first and second pipe loader assemblies **50** and **52** transport pipe sections **18**, shown in FIG. **3**, between the carriage **42** and the pipe box **54**.

(26) During operation, the carriage **42** uses a rotating spindle **56** and the wrenches **46** to connect pipe sections **18** to or remove pipe sections **18** from the drill string **16**. The carriage **42** moves longitudinally along a rail **58** positioned along the drill frame **36** to push and pull the drill string **16** through the ground surface **12**.

(27) With reference to FIGS. **4** and **5**, the first and second pipe loader assemblies **50** and **52** are each supported on the drill frame **36** such that they are parallel and spaced apart from one another. The first pipe loader assembly **50** is positioned adjacent the carriage **42** and the second pipe loader assembly **52** is positioned adjacent the wrenches **46**.

(28) The first pipe loader assembly **50** comprises a first shuttle arm **60** and a first lift assembly **62** supported on a first pipe loader frame **64**. The first pipe loader frame **64** comprises a front support **66** and a rear support **68**. Such supports **66** and **68** are positioned parallel to the drill frame **36** and are joined at a first end of the frame **64** by a bracket **70**. The supports **66** and **68** are joined at a second end of the frame **64** by the first lift assembly **62**.

(29) The second pipe loader assembly **52** comprises a second shuttle arm **72** and a second lift assembly **74** supported on a second pipe loader frame **76**. The second pipe loader frame **76** comprises a front support **78** and a rear support **80**. Such supports **78** and **80** are positioned parallel to the drill frame **36** and are joined at a first end of the frame **76** by the second lift assembly **74**. The supports **78** and **80** are joined at a second end of the frame **76** by a bracket **82**.

(30) The lift assemblies **62** and **74** are configured to move pipe sections **18** between the pipe box **54** and the shuttle arms **60** and **72**. The shuttle arms **60** and **72** are configured to move pipe sections **18** between the carriage **42** and the lift assemblies **62** and **74**.

(31) With reference to FIGS. **5-7**, each of the first and second pipe loader frames **64** and **76** is attached to the drill frame **36** by a mount **84**. Each mount **84** comprises a top plate **86** attached to an arm **88**. The arms **88** are each attached to the drill frame **36** and project from the side of the drill frame **36**, as shown in FIG. **4**. The top plate **86** is attached to the projecting end of each of the arm **88**. Each of the pipe loader frames **64** and **76** is supported on one of the top plates **86**, as shown in FIG. **7**.

(32) Turning back to FIG. **5**, the pipe box **54** is supported on each of the pipe loader assemblies **50** and **52**. The pipe box **54** attaches to each of the brackets **70** and **82** such that it is suspended above the shuttle arms **60** and **72** and the lift assemblies **62** and **74**. A plurality of dividers **90** are positioned at opposite ends of the interior of the pipe box **54**, as shown in FIG. **3**. The dividers **90** create columns within the pipe box **54** for storage of the pipe sections **18**. The pipe box **54** shown in FIGS. **2**, **3**, and **5** includes three columns. In alternative embodiments, the pipe box may include more than three columns or less than three columns.

(33) Continuing with FIGS. **5-7**, the mounts **84** of each pipe loader frame **64** and **76** are attached to the drill frame **36** by multiple welds. In alternative embodiments, the mounts may be attached to the drill frame with bolts, spring loaded pins, or the like, allowing the mounts to be selectively positioned along the length of the drill frame. Selectively positioning the mounts along the frame allows the drilling machine to be modified to accommodate different sizes of pipe sections. For example, if the drilling machine is originally configured for use with a pipe box sized to store 20-foot pipe sections, the mounts may be moved closer together so as to accommodate a pipe box

sized to store 15-foot pipe sections. The drilling machine may be configured so as to operate with various sizes of pipe sections.

(34) With reference to FIG. 8, each of the shuttle arms **60** and **72** comprises an elongate body **92** having a gripper **94** formed at its forward end **96**. The gripper **94** comprises an arm **98** configured to move towards and away from the body **92**. The gripper **94** is configured to releasably hold a pipe section **18** via movement of the arm **98**. Each shuttle arm **60** and **72** further comprises a shuttle pad **100** attached to its upper side **102** and extending along its length. The shuttle pads **100** provide a surface to support pipe sections **18** that are lowered from the pipe box **54** by the lift assemblies **62** and **74**.

(35) With reference to FIGS. 9 and 10, the shuttle arms **60** and **72** are moved using an actuator **104**. The actuator **104** shown in FIG. 9 comprises a rack **106** and a pinion gear **108** powered by a hydraulic motor **110**. In alternative embodiments, the actuator may comprise a hydraulic cylinder. Each pinion gear **108** is mounted on each pipe loader frame **64** and **76** beneath its corresponding shuttle arm **60** and **72**.

(36) Each pinion gear **108** and hydraulic motor **110** are supported by a set of brackets **118**, which are in turn supported on their corresponding pipe loader frame **64** and **76**. The brackets **118** further support a set of guides **122** positioned on opposite sides of the shuttle arms **60** and **72**, as shown in FIG. 11. The guides **122** secure each shuttle arm **60** and **72** to its corresponding pipe loader frame **64** and **76**.

(37) Turning back to FIG. 8, each of the shuttle arms **60** and **72** includes the rack **106**, which is an elongate metal structure either formed in or attached to a lower side **112** of each shuttle arm **60** and **72**. Each rack **106** extends between forward and rearward ends **96** and **114**, and preferably extends along the greater part of the length of its associated shuttle arm **60** and **72**, as shown in FIGS. 9 and 10. A plurality of longitudinally aligned grooves **116** are formed in the underside of each rack **106**.

(38) Turning back to FIGS. 9 and 10, a plurality of teeth **120** are formed around the periphery of each pinion gear **108**. The grooves **116** of each rack **106** mate with the teeth **120** of each pinion gear **108**. Rotation of each pinion gear **108** causes each shuttle arm **60** and **72** to move longitudinally relative to its corresponding pipe loader frame **64** and **76**. Rotation of each pinion gear **108** is driven by its corresponding hydraulic motor **110**.

(39) The pinion gears **108** may rotate in a clockwise or counter-clockwise direction. Clockwise rotation of the pinion gears **108** moves the shuttle arms **60** and **72** rearwardly away from the carriage **42**. Counter-clockwise rotation of the pinion gears **108** moves the shuttle arms **60** and **72** forward towards the carriage **42**.

(40) Turning back to FIG. 10, each of the shuttle arms **60** and **72** includes a set of front stops **124** and a rear stop **126**. The front stops **124** are formed on the lower side **112** of each shuttle arm **60** and **72** and comprise two tabs positioned on opposite sides of the rack **106**. The front stops **124** are configured to engage with ledges (not shown) formed at a rear end of the guides **122**. The front stops **124** engage with the ledges as the shuttle arms **60** and **72** move rearwardly and stop movement of the shuttle arms **60** and **72** beneath the third or last column of the pipe box **54**.

(41) The rear stop **126** is a tab attached to the rearward end **114** of the shuttle arms **60** and **72**. The rear stop **126** is configured to engage with a notch **128** formed on the set of brackets **118** as the shuttle arms **60** and **72** are moved forward towards the carriage **42**. Such engagement stops movement of the shuttle arms **60** and **72** once each shuttle arm's gripper **94** is aligned with the spindle **56**.

(42) In operation, the first shuttle arm **60** moves between its front and rear stops **124** and **126** along a first shuttle path. Likewise, the second shuttle arm **72** moves between its front and rear stops **124** and **126** along a second shuttle path. Both paths are transverse to the longitudinal axis of the first and second pipe loader frames **64** and **76** and the longitudinal axis **40** of the drill frame **36**.

(43) Turning back to FIG. 8, each shuttle arm **60** and **72** further includes a first stop **130** and a second stop **132**. Such stops **130** and **132** comprise a stepped tab attached to the side of each of the

shuttle arms **60** and **72**. The stops **130** and **132** are configured to engage with a vertically adjustable bolt **134**. The bolt **134** may comprise a flat plate joined to an elongate arm. Engagement of the bolt **134** with the first stop **130** stops movement of the shuttle arms **60** and **72** beneath the first column of the pipe box **54**. Engagement of the bolt **134** with the second stop **132** stops movement of the shuttle arms **60** and **72** beneath the second column of the pipe box **54**. In alternative embodiments, the shuttle arms may include more or less stops, depending on the number of columns included in the pipe box.

(44) Continuing with FIGS. **10** and **11**, the first and second lift assemblies **62** and **74** each comprise an arm **136** pivotally attached to two sets of brackets **138** via a pin **142**. The pin **142** and the brackets **138** join the front and rear supports **66** and **68** or **78** and **80** of the corresponding pipe loader frame **64** or **76**. A first end **140** of the arm **136** is pivotally attached to the pin **142** and brackets **138**, and a second end **144** of the arm **136** is positioned adjacent its corresponding shuttle arm **60** or **72**. A roller **146** is attached to the second end **144** of the arm **136**. The width of the roller **146** corresponds with the width of the pipe box **54**. The roller **146** supports the pipe sections **18** as they are transported between the pipe box **54** and the shuttle arms **60** and **72**.

(45) The first and second lift assemblies **62** and **74** each further comprise a hydraulic cylinder **148**. A first end **150** of the hydraulic cylinder **148** is attached to the brackets **138** and a second end **152** is attached to the lower side of the arm **138**. Extension and retraction of the hydraulic cylinder **148** raises and lowers the arm **138**. The hydraulic cylinder **148** includes a sensor configured to track the position of the cylinder's piston during operation. Thus, the hydraulic cylinder may be referred to as a “smart cylinder”. The sensor may communicate with a controller or processor located at the drilling machine's operator station **32**.

(46) The hydraulic cylinders **148** raise and lower the arms **138** in a radial motion. Thus, the lift assemblies **62** and **74** are considered “radial lift assemblies”. In alternative embodiments, the pipe loader assemblies may use vertical lift assemblies, like those described in U.S. Patent Publication No. 2019/0234158, authored by Porter et al. The size of the lift assemblies may vary depending on the size of the drilling machine, pipe box, and pipe sections.

(47) Turning back to FIG. **3**, to unload pipe sections **18** from the pipe box **54**, the lift assemblies **62** and **74** are initially in the raised position, holding the pipe sections **18** within the pipe box **54**. The shuttle arms **60** and **72** are positioned so that each of the grippers **94** is directly beneath the first column of the pipe box **54**. Once the grippers **94** are in position, the lift assemblies **62** and **74** are moved to a lowered position. The pipe sections **18** in the pipe box **54** will lower with the lift assemblies **62** and **74**. The lift assemblies **62** and **74** move lower than the height of the shuttle arms **60** and **72** when moving to the lowered position. Thus, the path of travel of the pipe sections **18** is interrupted by the shuttle arms **60** and **72** as the lift assemblies **62** and **74** lower. Such interruption causes the pipe section **18** from the first column to lower into the grippers **94** and the pipe sections **18** from the second and third columns to rest on the shuttle pads **100**.

(48) Once a pipe section **18** is securely held in the grippers **94**, the shuttle arms **60** and **72** will move slightly forward so the grippers **94** clear a front edge of the lift assemblies **62** and **74**. The shuttle arms **60** and **72** will slide underneath the pipe sections **18** resting on the shuttle pads **100** as the shuttle arms **60** and **72** move forward. A bottom edge of the pipe box **54** will prevent the pipe sections **18** resting on the shuttle pads **100** from moving with the shuttle arms **60** and **72**. Once the grippers **94** holding the pipe section **18** have cleared the lift assemblies **62** and **74**, the lift assemblies **62** and **74** will move to their raised positions. Pipe sections **18** remaining within the pipe box **54** are raised into the pipe box **54** as the lift assemblies **62** and **74** are raised.

(49) When unloading pipe sections **18** from the pipe box **54**, the first column must be completely unloaded before moving to the second column, and so on. Otherwise, pipe sections **18** would fall from the pipe box **54** as the lift assemblies **62** and **72** move to the lowered position.

(50) To load pipe sections **18** into the pipe box **54**, the lift assemblies **62** and **74** are initially in a lowered position. The shuttle arms **60** and **72** retrieve a pipe section **18** from the carriage **42** and

move rearwardly so that the grippers **94** are positioned directly beneath the third column. Once the pipe section **18** is directly beneath the third column of the pipe box **54**, the lift assemblies **62** and **74** will move to a raised position and pick up the pipe sections **18** along the way. The shuttle arms **60** and **72** will then move forward and retrieve another pipe section **18** from the carriage **42**.

(51) Once a new pipe section **18** is in the grippers **94**, the lift assemblies **62** and **74** will move to a lowered position so that the pipe section **18** within the third column will rest on the shuttle pads **100**. The shuttle arms **60** and **72** will then move rearwardly, sliding underneath the pipe section **18** resting on the shuttle pads **100**. Once the grippers **94** reach a position beneath the third column of the pipe box **54**, the pipe section **18** on the shuttle pads **100** will fall on top of the pipe section **18** held within the grippers **94**. The lift assemblies **62** and **74** are then moved to a raised position, lifting both of the pipe sections **18** into the third column of the pipe box **54**. The shuttle arms **60** and **72** may then move forward to retrieve another pipe section **18** from the carriage **42**. This process continues until the third column of the pipe box **54** is full of pipe sections **18**.

(52) When loading pipe sections **18** into the pipe box **54**, the third or last column must be completely filled before moving to the second column, and so on. Otherwise, pipe sections **18** would fall from the pipe box **54** as the lift assemblies **62** and **74** move to a lowered position.

(53) Continuing with FIGS. **2** and **3**, in operation, it is important that the shuttle arms **60** and **72** operate in unison when transporting a pipe section **18**. The pinion gears used with traditional shuttle arms are interconnected by a shaft so that the gears operate in unison. However, the shaft used to interconnect the gears is typically heavy and adds extra weight to the drilling machine.

(54) The drilling machine **10** shown in FIGS. **2** and **3** does not have a shaft interconnecting the pinion gears **108**. Thus, the pinion gears **108** are not mechanically coupled, apart from a pipe section **18** extending between the shuttle arms **60** and **72**. Not having a shaft extending between the pinion gears **108** removes excess weight from the drilling machine **10** and provides more space for other components, such as a tool box or fuel tank. As described below, the drilling machine **10** is configured so that the first and second shuttle arms **60** and **72** operate in unison without the use of a shaft interconnecting the pinion gears **108**.

(55) Turning back to FIGS. **8**, **9** and **12**, a first and second sensor **160** and **162** are used to track the position of the shuttle arms **60** and **72** along the first and second shuttle path. Parameters measured by the sensors **160** and **162** are transmitted to a controller. The controller analyzes the received parameters and directs operation of the actuators **104** in order to keep the shuttle arms **60** and **72** aligned as they move along their shuttle paths. The controller may comprise a computer processor supported at the drilling machine's operator station **32**. Alternatively, the controller may comprise a computer processor positioned remote from the drilling machine **10**.

(56) The first sensor **160** is attached to the brackets **118** opposite the hydraulic motor **110** on the first pipe loader frame **64**, as shown in FIGS. **8** and **9**. Likewise, the second sensor **162** is attached to the brackets **118** opposite the hydraulic motor **110** on the second pipe loader frame **76**, as shown in FIG. **12**. The first sensor **160** periodically measures a first parameter of the first shuttle arm **60**, while the second sensor **162** periodically measures a second parameter of the second shuttle arm **72**. The first and second parameters measured may be the position of the first and second shuttle arm **60** and **72** along their shuttle paths. Alternatively, the first and second parameters may be a parameter from which the position of the first and second shuttle arm **60** and **72** along their shuttle paths may be calculated.

(57) Continuing with FIGS. **8**, **9** and **12** each of the first and second sensors **160** and **162** comprises a non-contact absolute rotary encoder. During operation, the encoders track the position of the shuttle arms **60** and **72** relative to their respective pinion gears **108**. The encoders apply a value to various positions of the shuttle arms **60** and **72** along their shuttle paths. The encoders operate without the need for a reference point to recalibrate the encoder. The encoders are considered non-contact because they do not directly engage the pinion gears **108** or shuttle arms **60** and **72**. The absolute rotary encoder may comprise a magnetic, optical, or other type of non-contact encoder

known in the art.

(58) Turning to FIGS. **13** and **14**, an alternative embodiment of a sensor **164** is shown. The sensor **164** may be used in place of the non-contact sensors **160** or **162**. The sensor **164** comprise a contact absolute rotary encoder. The sensor **164** is considered a contact encoder because it is directly engaged to the pinion gear **108**. Like the sensors **160** and **162**, the sensor **164** applies a value to various positions of the shuttle arms **60** and **72** along their shuttle paths. In alternative embodiments, the sensor may comprise any form of a contact or mechanical rotary encoder known in the art.

(59) In an alternative embodiment, an incremental encoder may be used rather than an absolute rotary encoder. The incremental encoder may be used in conjunction with a proximity sensor. The proximity sensor may serve as a reference point for calibrating the incremental encoder.

(60) In further alternative embodiments, the first and second sensors may each comprise a camera, such as a video or time of flight camera. Such camera may directly view the shuttle arms and measure the position of the first shuttle arms along their shuttle paths. In even further alternative embodiments, any type of sensor capable of determining the position of the shuttle arms along their shuttle paths may be used.

(61) As the shuttle arms **60** and **72** move during operation, the sensors **160** and **162** continuously send measured parameters to the controller. Using the received parameters, the controller continually compares the position of the first shuttle arm **60** to the position of the second shuttle arm **72** to determine if the shuttle arms **60** and **72** are misaligned. Misalignment typically occurs if one shuttle arm **60** or **72** is moving faster than the other.

(62) One shuttle arm **60** or **72** may move slower than the other shuttle arm, because such shuttle arm experiences more resistance. For example, the angle at which the drill frame **36** is tilted about one or more of its axes may vary the amount of resistance encountered by each shuttle arm **60** and **72**. Typically, the drill frame **36** will be tilted at an angle so that the second pipe loader assembly **52** is lower than the first pipe loader assembly **50**, as shown in FIG. **2**. As a result, the second shuttle arm **72** may carry more of a pipe section's weight than the first shuttle arm **60**, leading to more resistance applied to the second shuttle arm **72** than the first shuttle arm **60**.

(63) Because misalignment is typically a result of one shuttle arm **60** or **72** moving faster than the other, the controller is configured to calculate a velocity at which each shuttle arm **60** and **72** is moving using the received parameters. In order to re-align the shuttle arms **60** and **72**, the controller may change the velocity at which one of the shuttle arms **60** and **72** is moving. The controller may control the velocity of each shuttle arm **60** and **72** by varying the flow rate of hydraulic fluid delivered to each hydraulic motor **110**. For such reason, each hydraulic motor **110** may utilize its own hydraulic circuit. Over time, the controller may learn the optimal flow rate to send to each hydraulic motor **110** to keep the shuttle arms **60** and **72** aligned.

(64) With reference to FIG. **15**, a method **200** of handling misalignment is shown. The method **200** involves realigning the shuttle arms **60** and **72** once they become misaligned. To start, the first and second shuttle arms **60** and **72** are moved, as shown by step **202**. The sensors **160** and **162** measure a first and second parameter for the shuttle arms **60** and **72**, as shown by step **204**. The measured parameters are transmitted to the controller for comparison, as shown by step **206**.

(65) If the shuttle arms **60** and **72** are determined to be aligned, the process will continue until the shuttle arms **60** and **72** reach their stopping position, as shown by steps **208** and **210**. If the shuttle arms **60** and **72** are determined to be misaligned, the controller will determine the velocity at which each shuttle arm **60** and **72** is moving. The controller will then direct the faster moving shuttle arm **60** or **72** to slow down until the slower moving shuttle arm **60** or **72** catches up, as shown by step **212**.

(66) The faster moving shuttle arm **60** or **72** is instructed to slow down because the shuttle arms are typically moving at full speed. However, if the shuttle arms **60** and **72** are not moving at full speed, the controller may instruct the slower moving shuttle arm **60** or **72** to speed up to catch the faster

moving shuttle arm. Such process will continue until the shuttle arms **60** and **72** reach their desired position, as shown by step **214**.

(67) With reference to FIG. **16**, another method **300** of handling misalignment of the shuttle arms **60** and **72** is shown. The goal of the method **300** is to prevent the shuttle arms **60** and **72** from becoming misaligned, rather than correcting misalignment on the fly. Such goal is accomplished using dynamic feedback.

(68) During operation, the controller can detect areas where one of the shuttle arms **60** or **72** may continually encounter resistance. Such resistance is detected by determining the velocity of each of the first and second shuttle arms **60** and **72** at successive positions along their respective shuttle paths. If one of the shuttle arms **60** or **72** moves slower than the other shuttle arm **60** or **72** through a certain segment of its shuttle path, the velocity of the faster moving shuttle arm is decreased within that segment. Alternatively, the velocity of the slower moving shuttle arm **60** or **72** may be increased within that segment.

(69) To start, the first shuttle arm **62** performs a first traverse of a first segment of the first shuttle path, as shown by step **302**. Simultaneously, the second shuttle arm **72** performs a first traverse of a first segment of the second shuttle path, as shown by step **302**. The parameters measured by the sensors **160** and **162** during movement of the shuttle arms **60** and **72** are transmitted to the controller for analysis, as shown by step **304**. The controller compares the velocity at which the first shuttle arm **60** traversed the first segment of the first shuttle path to the velocity at which the second shuttle arm **72** traversed the first segment of the second shuttle path, as shown by step **306**. Based on such comparison, the controller computes desired velocities for each shuttle arm **60** and **72** to traverse the first segment of each shuttle path so that the shuttle arms **60** and **72** stay aligned, as shown by step **308**.

(70) The controller directs the actuators **104** to move the shuttle arms **60** and **72** at the computed velocities each time the shuttle arms **60** and **72** traverse the first segment of their respective shuttle paths, as shown by steps **310**, **312**, and **314**. The sensors **160** and **162** continually measure parameters related to the position of the shuttle arms **60** and **72** each time the shuttle arms **60** and **72** traverse the first segment of their respective paths, as shown by step **316**. If the controller determines that the shuttle arms **60** and **72** are ever misaligned, the controller will calculate new velocities for each shuttle arm **60** and **72** to move at through the first segment of their respective shuttle paths, as shown by steps **318**, **320**, **322**, and **324**. Such process will continue throughout the drilling operation.

(71) The segments of the shuttle paths analyzed using the method **300** may be referred to as calibration zones. The controller may be configured to analyze and calculate desired velocities for the shuttle arms **60** and **72** to move at for multiple calibration zones throughout the shuttle paths. The calibration zones may correspond to the paths traveled by the shuttle arms **60** and **72** when loading or unloading pipe sections **18** from each column of the pipe box **54**.

(72) For example, when unloading pipe sections **18** from the pipe box **54**, a first calibration zone may comprise forward movement of the shuttle arms **60** and **72** from the first column of the pipe box **54** to the carriage **42**. A second calibration zone may comprise forward movement of the shuttle arms **60** and **72** from the second column of the pipe box **54** to the carriage **42**, and so on.

(73) When loading pipe sections into the pipe box **54**, a first calibration zone may comprise rearward movement of the shuttle arms **60** and **72** from the carriage **42** to the third column of the pipe box **54**. A second calibration zone may comprise rearward movement of the shuttle arms **60** and **72** from the carriage **42** to the second column of the pipe box **54**, and so on.

(74) The controller may pick which zones to analyze along the shuttle paths. Alternatively, an operator may set the zones for the controller. The first shuttle arm **60** may move at a different velocity in the first calibration zone as compared to the second calibration zone. Likewise, the second shuttle arm **72** may move at a different velocity through the first calibration as compared to the second calibration zone. The first shuttle arm **60**, for example, may also move at a different

velocity from the second shuttle arm **72** through the first calibration zone.

(75) As discussed, the controller will continually analyze parameters received by the sensors **160** and **162** throughout the drilling operation. It may be necessary to continually recalibrate the velocity of the shuttle arms **60** and **72** within each calibration zone because the resistance applied to each shuttle arm **60** and **72** may vary throughout operation. For example, some pipe sections **18** may be positioned differently within the shuttle arms **60** and **72** or some pipe sections **18** may contain more mud than others, causing the pipe sections **18** to vary in weight. Alternatively, the angle of the pipe box **54** may vary over the course of the drilling operation. In alternative embodiments, the controller may average a series of recorded velocities for each calibration zones and instruct the actuators to move the shuttle arms at the average velocity for each calibration zone.

(76) The calibration zones are only needed for those times when the shuttle arms **60** and **72** are carrying a pipe section **18**. If the shuttle arms **60** and **72** are moving to a position to retrieve a pipe section **18**, it is not necessary that the arms move in unison. As such, the first and second shuttle arms **60** and **72** may intentionally be moved at different speeds and times from one another.

(77) In operation, the hydraulic motors **110** used to drive rotation of each pinion gear **108** use the same hydraulic pump. Thus, a shuttle arm **60** or **72** moves faster by itself, as compared to moving the shuttle arms **60** and **72** at the same time. As such, there may be instances where the drilling process can be made more efficient if the shuttle arms **60** and **72** are moved at different times.

(78) With reference to FIG. **17**, a method **400** of operating the shuttle arms **60** and **72** independently while adding pipe sections **18** to the drill string **16** is shown. To start, the shuttle arms **60** and **72** deliver a pipe section **18** to the carriage **42**, as shown by step **402**. After the pipe section **18** is attached to the spindle **56**, the first shuttle arm **60** may move rearward back to the pipe box **54**, as shown by steps **404** and **406**. Once the first shuttle arm **60** is out of the way of the carriage **42**, the carriage **42** may move forward along the rail **58**, as shown by step **408**. The second shuttle arm **72** may start to move rearwardly once the first shuttle arm **60** is out of the way of the carriage **42**, as shown by step **410**. Alternatively, the second shuttle arm **72** may loosely grip the pipe section **18** as the carriage **42** moves forward to help guide the pipe section **18** towards the drill string **16**.

(79) Turning to FIG. **18**, a method **500** of operating the shuttle arms **60** and **72** independently while removing pipe sections **18** from the drill string **16** is shown. To start, the carriage **42** pulls the drill string **16** from the ground surface **12**, as shown by step **502**. Once the carriage **42** passes the second shuttle arm **72**, the second shuttle arm **72** moves forward towards the drill frame **36** and holds the pipe section **18**, as shown by steps **504** and **506**. Likewise, once the carriage **42** passes the first shuttle arm **60**, the first shuttle arm **60** moves forward towards the drill frame **36** and holds the pipe section **18**, as shown by steps **508** and **501**. After the wrenches **46** and spindle **56** remove the pipe section **18** from the drill string **16**, the shuttle arms **60** and **72** grip the pipe section **18** and transport it to the pipe box **54** as shown by step **512**. To save time, the wrenches **46** may unthread the pipe section **18** from the drill string **16** as only second shuttle arm **72** is holding the pipe section **18**.

(80) The shuttle arms **60** and **72** may also be configured so that they are selectively movable. The controller may include a user interface that allows an operator to independently move each shuttle arm **60** and **72** to a desired position at any time. For example, only one shuttle arm **60** or **72** may be moved forward towards the carriage **42** to hold a tool or a small pipe section.

(81) The shuttle arms **60** and **72** may be configured to automatically move slower once the gripper **94** on each arm starts to move beneath the pipe box **54**. The slower movement gives the operator time to change which column the shuttle arm **60** or **72** is moving towards, if needed.

(82) The shuttle arms **60** or **72** may also be moved independently to help prepare the drilling machine **10** for transport. When transporting the drilling machine **10**, it is beneficial to position the carriage **42** midway along the drill frame **36** in order to help balance the drilling machine **10**. Such position of the carriage **42** may be referred to as a “transport position”.

(83) With reference to FIG. **19**, a method **600** of operating the shuttle arms **60** and **72**

independently in order to move the carriage **42** to the transport position is shown. To start, the drilling operator may activate transport mode, as shown by step **602**. Transport mode may be activated on a user interface located at the operator station **32**. Once activated, the controller determines where the carriage **42** is located along the drill frame **36**, as shown by step **604**.

(84) If the carriage **42** is behind the transport position, the controller retracts the first shuttle **60** and extends the second shuttle **72**, as shown by step **606**. The carriage **42** then moves forward along the drill frame **36** to the transport position, as shown by step **608**. Once the carriage **42** is at the transport position, the first shuttle arm **60** may extend, as shown by step **610**. Following step **610**, the controller notifies the drilling operator that carriage **42** is ready for transport, as shown by step **618**.

(85) If the carriage **42** is in front of the transport position, the controller retracts the second shuttle arm **72** and extends the first shuttle arm **60**, as shown by step **612**. The carriage **42** then moves rearward along the drill frame **36** to the transport position, as shown by step **614**. Once the carriage **42** is at the transport position, the second shuttle arm **72** may extend, as shown by step **616**. Following step **616**, the controller notifies the drilling operator that carriage **42** is ready for transport, as shown by step **618**.

(86) Because the shuttle arms **60** and **72** can move independently, the arms **60** and **72** may also be used as weights to balance the drilling machine **10** during transport. For example, one shuttle arm **60** or **72** may be extended towards the carriage **42** while the other shuttle arm **60** or **72** is positioned beneath the pipe box **54**.

(87) Changes may be made in the construction, operation and arrangement of the various parts, elements, steps and procedures described herein without departing from the spirit and scope of the invention as described in the following claims.

Claims

1. An apparatus, comprising: an elongate frame having a longitudinal axis; a first shuttle arm supported by the frame and movable along a first shuttle path transverse to the longitudinal axis of the frame; a second shuttle arm supported by the frame and movable along a second shuttle path spaced from, but parallel to, the first shuttle path; a first actuator configured to power movement of the first shuttle arm along the first shuttle path; and a second actuator configured to power movement of the second shuttle arm along the second shuttle path; in which each of the first and second actuators comprises a pinion; and in which the pinion of the first actuator is not connected to the pinion of the second actuator, apart from any removable load transported by both the first and second shuttle arms.
2. The apparatus of claim 1, further comprising: a first sensor that periodically measures a first parameter that is either the position of the first shuttle arm or a parameter from which such position may be calculated.
3. The apparatus of claim 2, further comprising: a second sensor that periodically measures a second parameter that is either the position of the second shuttle arm or a parameter from which such position may be calculated.
4. The apparatus of claim 3, further comprising: a controller in communication with the first and second sensors and with the first and second actuators, the controller configured to evaluate the first and second parameters, and to issue commands to adjust a velocity of one of the shuttle arms in response to that evaluation.
5. The apparatus of claim 4, in which the first shuttle path is perpendicular to the longitudinal axis of the frame; and in which the controller is configured to command the first actuator to move the first shuttle arm at a first velocity and to command the second actuator to move the second shuttle arm at a second velocity, wherein the first velocity and the second velocity are different velocities.
6. The apparatus of claim 4, in which the controller is configured to command the first and second

shuttle arms to operate in unison.

7. The apparatus of claim 1, in which each of the first and second actuators further comprises a hydraulic motor used to power rotation of that actuator's pinion.

8. The apparatus of claim 3, in which each of the first and second sensors comprises an encoder.

9. The apparatus of claim 8, in which the encoder is a rotary encoder.

10. A horizontal boring machine, comprising: the apparatus of claim 1; and a carriage supported on the frame and movable between a first and second end of the frame.

11. The horizontal boring machine of claim 10, further comprising: a spindle supported on the carriage; and a pipe box supported on the frame; in which the first and second shuttle arms are movable between the pipe box and the spindle.

12. An apparatus, comprising: an elongate frame having a longitudinal frame axis; a first shuttle arm supported by the frame and movable along a first shuttle path by rotation of a first pinion, the first shuttle arm configured to support a first portion of a pipe section; and a second shuttle arm supported by the frame and movable along a second shuttle path spaced from, but parallel to, the first shuttle path by rotation of a second pinion, the second shuttle arm configured to support a second portion of the pipe section; in which the first pinion is separate from the second pinion such that rotation of the first pinion and rotation of the second pinion are independently adjustable.

13. The apparatus of claim 12 in which the first pinion and the second pinion are not interconnected by a shaft.

14. The apparatus of claim 12 further comprising: a first sensor configured to measure a first parameter of the first shuttle arm; a second sensor configured to measure a second parameter of the second shuttle arm; and a controller configured to independently adjust the rotation of the first pinion and the second pinion in response to the first parameter and the second parameter.
